Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/26/2005
PI Name:	Johnson, William Ph.D.		
Project Title:	Thermodynamic and Transport Properties of Glass Forming Alloy Liquids Using the Caltech High Vacuum Electrostatic Levitator Platform		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	91125	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Studies of the thermophysical properties of glass forming liquid metallic alloys are carried out using a High Vacuum Electro-static Levitation System as a platform. The Caltech HVESL system has been developed for this purpose. The system utilizes a 4-beam tetrahedral laser heating system to process liquid drops under high vacuum with very uniform drop temperature. We measure liquid heat capacity, viscosity, surface tension, and TTT-diagrams for crystallization. We are developing methods for measurement of atomic diffusion constants in liquid alloys based on the minimization of gravity driven and Marangoni convection. This is accomplished by processing the melt under very near isothermal and quiescent conditions to eliminate gravity driven and Marangoni convection effects on diffusion profiles. We have carried out measurements on a number of glass forming liquid alloys and have developed "process maps" and protocols for carrying out diffusion measurments. A comprehensive study and analysis of atomic diffusion measurements on the HVES has recently been published [J. Schroers, S. Bossuyt, W.K. Rhim, and W.L. Johnson, Rev. Sci. Instr., 75, 4523

Task Description:	(2004)] The objective of this task is to carry out accurate viscosity and atomic diffusion measurements on liquid metallic alloys both in the equilibrium and undercooled liquid states. The research will evaluate and characterize the effects of gravity on such measurements. Specifically, we wish to evaluate diffusion and viscosity in undercooled liquids as a function of temperature throughout the range from the melting point down to the glass transition. A primary objective is to understand the role of gravity in liquid diffusion studies and develop a "benchmark" micro-gravity experiment which effectively eliminates the influence of convection on diffusion profiles. This task establishes the HVELS platform as the preferred technique for characterizing liquid metals and alloys. The method provides "benchmark" data for viscosity, atomic diffusion, thermal conductivity, surface tension, density, and heat capacity on reactive liquids at high temperature. Data on the kinetics of crystallization are used to generate liquid "processing maps". These data are fundamental inputs for the development of commercial processes for casting and molding of net shape, high performance, amorphous metal hardware and products. The outputs of this project are being directly incorporated into commercial production of amorphous metal products by Liquidmetal Technologies Corp, Ti-Met Corp., and other companies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Data generated on the Caltech HVESL facility in this work has been directly used for design, simulation, and modeling of manufacturing processes used by Liquidmetal Technologies, Lake Forest, CA to produce metallic glass products. Data for liquid Ti-based alloys has been utilized by Ti-Met Corp., Henderson, NV.
Task Progress:	(1) We have designed and constructed a symmetric 4-beam laser heating system used to process liquids with very low temperature gradients. (2) We have determined processing an flow maps for rough 6 different families of bulk metallic glass forming allows. (3) Measurements of density, heat capacity, viscosity, surface tension, and crystallization kinetics have been carried out. The results have been documented in a series of 6-7 papers published in the archival literature. (4) A detailed method for carrying out atomic diffusion measurements on reactive liquids at high temperature has been developed, assessed, and tested Results have been documented in a published article in Reviews of Scientific Instrucments (2004).
Bibliography Type:	Description: (Last Updated: 07/06/2007)
Articles in Peer-reviewed Journals	Mukherjee S, Johnson WL, Rhim WK. "Noncontact measurement of high-temperature surface tension and viscosity of bulk metallic glass-forming alloys using the drop oscillation technique." Applied Physics Letters 2005 Jan 3;86(1):014104. http://dx.doi.org/10.1063/1.1844596 , Jan-2005
Articles in Peer-reviewed Journals	Mukherjee S, Kang HG, Johnson WL, Rhim WK. "Noncontact measurement of crystallization behavior, specific volume, and viscosity of bulk glass-forming Zr-Al-Co-(Cu) alloys." Physical Review B 2004 Nov;70(17):174205. http://dx.doi.org/10.1103/PhysRevB.70.174205 , Nov-2004
Articles in Peer-reviewed Journals	Schroers J, Bossuyt S, Rhim WK, Johnson WL. " Enhanced temperature uniformity by tetrahedral laser heating." Review of Scientific Instruments 2004 Nov;75(11):4523-7. http://dx.doi.org/10.1063/1.1804351 , Nov-2004
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Articles in Peer-reviewed Journals	Mukherjee S, Zhou Z, Schroers J, Rhim WK, Johnson WL. "Overheating threshold and its effect on time–temperature-transformation diagrams of zirconium based bulk metallic glasses." Applied Physics Letters 2004 Jun 14; 84(24):5010-2. http://dx.doi.org/10.1063/1.1763219 , Jun-2004
Articles in Peer-reviewed Journals	Mukherjee S, Zhou Z, Johnson WL, Rhim WK. " Thermophysical properties of Ni–Nb and Ni–Nb–Sn bulk metallic glass-forming melts by containerless electrostatic levitation processing." Journal of Non-Crystalline Solids 2004 Jun 1;337(1):21-8. http://dx.doi.org/10.1016/j.jnoncrysol.2004.03.102 , Jun-2004
Dissertations and Theses	Mukherjee S. "Thermophysical properties of bulk metallic glass forming liquids by High High Vacuum Electrostatic Levitation Processing." Ph.D. thesis, California Institute of Technology, Caltech Library electronic thesis files, June 2005. , Jun-2005